

Measure Theory

Labix

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Abstract

Contents

1	Measure Theory	3
1.1	σ -Algebra	3
1.2	Measures	3
1.3	Borel Measures	4
2	Measure Spaces	5
2.1	Measure Spaces	5
2.2	Measurable Functions	5
2.3	Convergence	5
3	Integration Theory	6
3.1	Integration of Measurable Functions	6
3.2	Properties of the Lebesgue Integral	6
3.3	Comparison to Riemann Integrability	7
3.4	The Space of Measurable Functions	7
4	Lebesgue Measure in $\mathbb{R}^n$	8
4.1	Lebesgue Measure	8
4.2	Lebesgue Integral	9
5	Differentiation	10
5.1	Existence of Anti-Derivatives	10

1 Measure Theory

1.1 σ -Algebra

Definition 1.1.1 (σ -algebra)

Let X be a set. Let $\mathcal{F} \subseteq \mathcal{P}(X)$. We say that $\mathcal{F}$ is a σ -algebra if the following are true.

- $S \in \mathcal{F}$.
- If $A \in \mathcal{F}$, then $X \setminus A \in \mathcal{F}$.
- If $A_k \in \mathcal{F}$ for $k \in \mathbb{N} \setminus \{0\}$, then $\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}$

We say that $A \in \mathcal{F}$ is a measurable set.

Lemma 1.1.2

Let X be a set. Let $\mathcal{F}$ a σ -algebra. Then the following are true.

- $\emptyset \in \mathcal{F}$.
- If $A_k \in \mathcal{F}$ for $k \in \mathbb{N}$, then $\bigcap_{k=0}^{\infty} A_k \in \mathcal{F}$.

Definition 1.1.3 (Smallest σ -algebra Containing a Set)

Let X be a set. Let $P \subseteq \mathcal{P}(X)$. Define the smallest σ -algebra containing P by $\sigma(P)$.

Lemma 1.1.4

Let X be a set. Let $P \subseteq \mathcal{P}(X)$ be a subset. Then we have

$$\sigma(P) = \bigcap_{\substack{\mathcal{F} \supseteq P \\ \mathcal{F} \text{ is measurable}}} \mathcal{F}$$

1.2 Measures

Definition 1.2.1 (Measure)

Let X be a set. Let $\mathcal{F}$ be a σ -algebra of X . Let $\mu : \mathcal{F} \rightarrow [0, \infty)$ be a function. We say that μ is a measure if the following are true.

- $\mu(\emptyset) = 0$
- If $A_1, \dots, A_k, \dots$ are pairwise disjoint in $\mathcal{F}$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k)$$

Proposition 1.2.2

Let X be a set. Let $\mathcal{F}$ be a σ -algebra of X . Let $\mu : \mathcal{F} \rightarrow [0, \infty)$ be a measure on X .

- If $A_1, A_2 \in \mathcal{F}$ and $A_1 \subseteq A_2$, then

$$\mu(A_1) \leq \mu(A_2)$$

- If $A_k \in \mathcal{F}$ for $k \in \mathbb{N} \setminus \{0\}$, then we have

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{\infty} \mu(A_k)$$

- For any $A_1, A_2 \in \mathcal{F}$, we have

$$\mu(A_1) + \mu(A_2) = \mu(A_1 \cup A_2) - \mu(A_1 \cap A_2)$$

Proposition 1.2.3

Let X be a set. Let $\mathcal{F}$ be a σ -algebra of X . Let $\mu : \mathcal{F} \rightarrow [0, \infty)$ be a measure on X . The the following are true.

- If $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_k \subseteq \cdots$ are measurable subsets, then we have

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \lim_{k \in \mathbb{N}} \mu(A_k)$$

- If $B_1 \supseteq B_2 \supseteq \cdots \supseteq B_k \supseteq \cdots$ are measurable subsets, then we have

$$\mu \left(\bigcap_{k=1}^{\infty} B_k \right) = \lim_{k \in \mathbb{N}} \mu(B_k)$$

Definition 1.2.4 (Outer Measures)

Let X be a set. Let $\nu : P(X) \rightarrow \emptyset$. We say that ν is an outer measure.

- $\nu(\emptyset) = 0$.
- If $A_1 \subseteq A_2$, then

$$\nu(A_1) \leq \nu(A_2)$$

- If $A_1, \dots, A_k, \dots$ are subsets, then

$$\nu \left(\bigcup_{k=1}^{\infty} A_k \right) \leq \sum_{k=1}^{\infty} \nu(A_k)$$

Lemma 1.2.5

Let X be a set. Let $\mathcal{F}$ be a σ -algebra. Let μ be a measure. Then μ is an outer measure.

1.3 Borel Measures

Definition 1.3.1 (Borel σ -algebra)

Let $(X, \mathcal{T})$ be a topological space. Define the Borel σ -algebra of X to be

$$\mathcal{B}(X) = \sigma(\mathcal{T})$$

Definition 1.3.2 (Borel Measure)

Let X be a topological space. A Borel measure is a measure $\mu : \mathcal{B}(X) \rightarrow [0, \infty)$ on X .

Definition 1.3.3 (Radon Measure)

Let X be a topological space. Let μ be a Borel measure. We say that X is Radon if for any compact subset $K \in \mathcal{B}(X)$, we have

$$\mu(K) < \infty$$

2 Measure Spaces

2.1 Measure Spaces

Definition 2.1.1 (Measurable Space)

Let X be a set. We say that X is measurable if there exists a σ -algebra $\mathcal{F}$ and a measure $\mu : \mathcal{F} \rightarrow [0, \infty)$ on X .

Definition 2.1.2 (Measure Space)

A measure space $(X, \mathcal{F}, \mu)$ consists of a set X , a σ -algebra $\mathcal{F}$ and a measure μ on X .

Definition 2.1.3 (Finiteness of Measure Spaces)

Let $(X, \mathcal{F}, \mu)$ be a measure space.

- We say that X is finite if $\mu(X) < \infty$.
- We say that X is σ -finite if there exists a collection $\{U_k \in \mathcal{F} \mid k \in \mathbb{N} \setminus \{0\}\}$ such that $X = \bigcup_{k=1}^{\infty} U_k$ and $\mu(U_k) < \infty$.

Lemma 2.1.4

Let $(X, \mathcal{F}, \mu)$ be a measure space. If X is finite, then X is σ -finite.

2.2 Measurable Functions

Definition 2.2.1 (Measurable Functions)

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. Let $f : E \rightarrow F$. We say that f is measurable if for all $A \in \mathcal{F}$, $f^{-1}(A) \in \mathcal{E}$.

Lemma 2.2.2

Let $(E, \mathcal{E})$, $(F, \mathcal{F})$ and $(G, \mathcal{G})$ be measurable spaces. Then the following are true.

- If $f : E \rightarrow F$ and $g : F \rightarrow G$ are measurable functions, then $g \circ f$ is measurable.
- $\text{id}_E : E \rightarrow E$ is measurable.

Definition 2.2.3 (Pushforward Measure)

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. Let $f : E \rightarrow F$ be a measurable function. Let $\mu : \mathcal{E} \rightarrow [0, \infty)$ be a measure. Define the push forward measure $\mu_* : \mathcal{F} \rightarrow [0, \infty)$ by

$$\mu_*(A) = \mu(f^{-1}(A))$$

2.3 Convergence

Definition 2.3.1 (Convergence Almost Everywhere)

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. Let $\mu : \mathcal{E} \rightarrow [0, \infty)$ be a measure. Let $(f_n : E \rightarrow F)_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence of measurable functions. We say that $(f_n)_{n \in \mathbb{N} \setminus \{0\}}$ converges almost everywhere to a measurable function $f : E \rightarrow F$ if

$$\mu(\{x \in E \mid (f_n(x))_{n \in \mathbb{N} \setminus \{0\}} \text{ does not converge to } f(x)\}) = 0$$

Definition 2.3.2 (Convergence in Measure)

Let $(E, \mathcal{E})$ be a measurable space. Let $\mu : \mathcal{E} \rightarrow [0, \infty)$ be a measure. Let $(f_n : E \rightarrow \mathbb{R})_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence of measurable functions. We say that $(f_n)_{n \in \mathbb{N} \setminus \{0\}}$ converges in measure to a measurable function $f : E \rightarrow \mathbb{R}$ if

$$\mu(\{x \in E \mid |f(x) - f_n(x)| > \varepsilon\}) \rightarrow 0$$

as $n \rightarrow \infty$.

3 Integration Theory

3.1 Integration of Measurable Functions

Definition 3.1.1 (Simple Functions)

Let $(E, \mathcal{E}, \mu)$ be a measure space. A simple function is a function of the form

$$f(x) = \sum_{k=1}^n a_k 1_{A_k}(x)$$

for $A_1, \dots, A_n$ disjoint measurable sets and $a_k \in [0, \infty)$.

Definition 3.1.2 (Lebesgue Integral for Simple Functions)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $f(x) = \sum_{k=1}^n a_k 1_{A_k}(x)$ be a simple function. Define the Lebesgue integral of f to be

$$\int f \, d\mu = \sum_{k=1}^n a_k \mu(A_k)$$

Lemma 3.1.3

Definition 3.1.4 (Lebesgue Integral for Positive Functions)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $f : E \rightarrow \mathbb{R}$ be a positive measurable function. Define the Lebesgue integral of f to be

$$\int f \, d\mu = \sup \left\{ \int g \, d\mu \mid g \text{ is a simple function and } g \leq f \right\}$$

Definition 3.1.5 (Lebesgue Integral for General Functions)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $f : E \rightarrow \mathbb{R}$ be a measurable function. Let f_+ be the positive part of f and let f_- be the negative part of f . Define the Lebesgue integral of f to be

$$\int f \, d\mu = \int f_+ \, d\mu + \int -f_- \, d\mu$$

3.2 Properties of the Lebesgue Integral

Theorem 3.2.1 (Monotone Convergence Theorem)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $f : E \rightarrow [0, \infty)$ be a non-negative measurable function. Let $(f_n : E \rightarrow [0, \infty))_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence of non-negative measurable functions. If $(f_n) \uparrow f$, then

$$\int f_n \, d\mu \uparrow \int f \, d\mu$$

Proposition 3.2.2 (Beppo-Levi)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $(f_n : E \rightarrow \mathbb{R})_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence of measurable functions. Then we have

$$\int \sum_n f_n \, d\mu = \sum_n \int f_n \, d\mu$$

Theorem 3.2.3 (Fatou's Lemma)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $(f_n : E \rightarrow [0, \infty))_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence of non-negative measurable functions. Then we have

$$\int \left(\liminf_n f_n \right) \, d\mu \leq \liminf_n \int f_n \, d\mu$$

Theorem 3.2.4 (Dominated Convergence Theorem)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $(f_n : E \rightarrow \mathbb{R})_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence of measurable functions. Let $f : E \rightarrow \mathbb{R}$ be a measurable function such that f_n converges to f almost everywhere. Suppose that there exists a positive function $g : E \rightarrow \mathbb{R}$ such that $|f| \leq g$ and $|f_n| \leq g$ for all n and $\int g < \infty$. Then we have

$$\lim_n \int f_n d\mu = \int f d\mu$$

3.3 Comparison to Riemann Integrability**3.4 The Space of Measurable Functions****Definition 3.4.1 (The L^p Space of a Measure Space)**

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $p \geq 1$. Define the associated L^p space of E to be the set of measurable functions

$$L^p(E) = \{f : E \rightarrow \mathbb{R} \mid f \text{ is measurable}\}$$

together with the norm function $\|\cdot\|_p : L^p(E) \rightarrow \mathbb{R}$ defined by

$$\|f\|_p = \left(\int |f|^p d\mu \right)^{1/p}$$

Lemma 3.4.2

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $p \geq 1$. Then $L^p(E)$ is a normed space.

Proposition 3.4.3 (Holder's Inequality)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $p, q \geq 1$ such that $1/p + 1/q = 1$. Let $f, g : E \rightarrow \mathbb{R}$ be measurable functions. Then we have

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

Proposition 3.4.4 (Minkowski's Inequality)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $p \geq 1$. Let $f, g : E \rightarrow \mathbb{R}$ be measurable functions. Then we have

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Proposition 3.4.5 (Markov's Inequality)

Let $(E, \mathcal{E}, \mu)$ be a measure space. Let $f : E \rightarrow [0, \infty)$ be a non-negative measurable function. Let $\lambda > 0$. Then we have

$$\mu(\{x \in E \mid f(x) > \lambda\}) \leq \frac{1}{\lambda} \int f d\mu$$

4 Lebesgue Measure in $\mathbb{R}^n$

4.1 Lebesgue Measure

Definition 4.1.1 (Lebesgue Outer Measure)

Define the Lebesgue outer measure to be the function $\lambda^* : \mathcal{P}(\mathbb{R}^n) \rightarrow [0, \infty]$ given by

$$\lambda^*(S) = \inf \left\{ \sum_{i=1}^{\infty} \text{Vol}(R_i) \mid R_i \text{ is a rectangle for } i \in \mathbb{N}^+ \text{ and } S \subseteq \bigcup_{i=1}^{\infty} R_i \right\}$$

Proposition 4.1.2

Let $S \subseteq \mathbb{R}^n$ be a bounded subset. Then the following are true.

- $\mu_*(S) \leq \lambda^*(S) \leq \mu^*(S)$.
- If S is Jordan measurable, then we have $\mu(S) = \lambda^*(S)$.

Definition 4.1.3 (Lebesgue Measurable Sets)

Let $S \subseteq \mathbb{R}^n$ be a subset. We say that S is Lebesgue measurable if for all $\varepsilon > 0$, there exists an open subset $U \subseteq \mathbb{R}^n$ such that $S \subseteq U$ and

$$\lambda^*(U \setminus S) < \varepsilon$$

In this case, we say that the Lebesgue measure of S is

$$\lambda(S) = \lambda^*(S)$$

Definition 4.1.4 (Set of Lebesgue Measurable Sets)

Define the set of Lebesgue measurable sets by

$$\mathcal{L}(\mathbb{R}^n) = \{S \subseteq \mathbb{R}^n \mid S \text{ is Lebesgue measurable}\}$$

Proposition 4.1.5

The following are true regarding Lebesgue measurability.

- $\emptyset$ is Lebesgue measurable.
- If $U \subseteq \mathbb{R}^n$ is Lebesgue measurable, then $\mathbb{R}^n \setminus U$ is Lebesgue measurable.
- If U_i is Lebesgue measurable for each $i \in \mathbb{N}^+$, then $\bigcup_{i=1}^{\infty} U_i$ is Lebesgue measurable.
- If U_i is Lebesgue measurable for each $i \in \mathbb{N}^+$, then $\bigcap_{i=1}^{\infty} U_i$ is Lebesgue measurable.

Proposition 4.1.6

The following are true regarding Lebesgue measurability.

- If $U \subseteq \mathbb{R}^n$ is open, then U is Lebesgue measurable.
- If $U \subseteq \mathbb{R}^n$ is closed, then U is Lebesgue measurable.
- If $U \subseteq \mathbb{R}^n$ is such that $\lambda^*(U) = 0$, then U is Lebesgue measurable.

Proposition 4.1.7

Let $S \subseteq \mathbb{R}^n$ be a subset. Then the following are equivalent.

- S is Lebesgue measurable.
- For all $\varepsilon > 0$, there exists an open subset $U \subseteq \mathbb{R}^n$ such that $S \subseteq U$ and $\lambda^*(U \setminus S) < \varepsilon$.
- For all $\varepsilon > 0$, there exists a closed subset $U \subseteq \mathbb{R}^n$ such that $S \subseteq U$ and $\lambda^*(U \setminus S) < \varepsilon$.

Lemma 4.1.8

Let $S \subseteq \mathbb{R}^n$ be a subset. If S is Jordan measurable, then S is Lebesgue measurable.

Theorem 4.1.9 (Monotone Convergence Theorem)

Let $S_i \subseteq \mathbb{R}^n$ be Lebesgue measurable for $i \in \mathbb{N}^+$. Then the following are true.

- If $S_i \subseteq S_{i+1}$ for each $i \in \mathbb{N}^+$, then we have

$$\lambda \left(\bigcup_{i=1}^{\infty} S_i \right) = \lim_{i \rightarrow \infty} \lambda(S_i)$$

- If $S_i \supseteq S_{i+1}$ for each $i \in \mathbb{N}^+$ and there exists $i \in \mathbb{N}^+$ such that $\lambda(S_i) < \infty$, then we have

$$\lambda \left(\bigcap_{i=1}^{\infty} S_i \right) = \lim_{i \rightarrow \infty} \lambda(S_i)$$

Theorem 4.1.10 (Caratheodory Criterion)

Let $S \subseteq \mathbb{R}^n$ be a subset. Then the following are equivalent.

- S is Lebesgue measurable.
- For all subsets $A \subseteq \mathbb{R}^n$, we have

$$\lambda^*(A) = \lambda^*(A \cap S) + \lambda^*(A \setminus S)$$

- For every elementary set $E \subseteq \mathbb{R}^n$, we have

$$\lambda^*(E) = \lambda^*(E \cap S) + \lambda^*(E \setminus S)$$

- For every rectangle $R \subseteq \mathbb{R}^n$, we have

$$\lambda^*(R) = \lambda^*(R \cap S) + \lambda^*(R \setminus S)$$

Proposition 4.1.11

Let $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$ be Lebesgue measurable. Then $S \times T$ is Lebesgue measurable and

$$\lambda(S \times T) = \lambda(S) \cdot \lambda(T)$$

4.2 Lebesgue Integral

5 Differentiation

5.1 Existence of Anti-Derivatives